Independent audit of one-loop charges and static defect absorption

Project audit for Gustav Delius

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Abstract

The proposed collective-coordinate momentum is an exact derivative of the proposed one-loop Lagrangian; a regulated Gaussian mode sum supports that Lagrangian conditionally. **(c)** Staticity does not justify changing which bulk soliton charge enters absorption. The defect's nonlinear boundary potential and localized charge operators require a renormalization argument beyond the bulk tadpole, which vanishes on its constant vacua. **(d)** A shift of the absorption locus reconciles both charge pairs algebraically **(c)**, but no quantum sewing calculation establishes it physically **(d)**. Existing good-equation conventions and bulk subtraction are retained; no new renormalization prescription is chosen.

1 Scope, labels, and evidence

Labels are **(a)** literature results, **(b)** literature heuristics, **(c)** project derivations, and **(d)** unestablished physical identifications or possibilities. Every displayed new **(c)** identity is checked independently in `phase2/scripts/audit_charges_action.py`, using SymPy and 65-digit mpmath. The script imports no existing project charge formulas.

The main audited passages are `charge_frame_master.tex`, lines 284–316 (co-moving action), 325–363 (derivative), and 367–398 (static-defect interpretation); also `oneloop_defect_phase.tex`, lines 605–650 (renormalization and absorption). The previous independent Phase 2h audit was consulted as a lead; the checks here were implemented separately.

2 What the co-moving mode sum does establish

Write the potential fluctuation in $\xi = x - vt$ as φ, and let $u_0 = \partial_\xi \phi_0$. Taylor expansion of the fixed project action gives **(c)** (`audit_charges_action.py`)

$$\mathcal{L}_2 = \tfrac{1}{2}(\varphi_t - v\varphi_\xi)^2 - \tfrac{1}{2}(1 - 6u_0)\varphi_\xi^2 - \tfrac{1}{2}\varphi_{\xi\xi}^2. \tag{1}$$

Thus asymptotic continuum modes have co-moving frequency $\Omega(q) = |q|\sqrt{1+q^2} - vq$. This follows by substitution of a plane wave into the vacuum quadratic action, and is positive for $q \neq 0$, $|v| < 1$. The determinant interpretation still requires a complete, correctly normalized mode basis, the collective-coordinate projection, and the specified boundary conditions. This audit does not supply those parts. **(d)**

There is nevertheless a useful conditional derivation beyond checking a Legendre identity. If equal-mode-count Gaussian quantization is valid, its continuum contribution uses the derivative form of the spectral trace, including its cutoff endpoint. For $\theta(q) = \sum_r c_r \Theta(q; \phi_r)$ on the existing

soliton branches, **(c)** (`audit_charges_action.py`):

$$E_{\rm dens}(\Lambda) = \frac{1}{4\pi}\int_{-\Lambda}^{\Lambda} |q|\sqrt{1+q^2}\,\theta'(q)\,dq, \tag{2}$$

$$B(\Lambda) = \frac{\Lambda\sqrt{1+\Lambda^2}}{4\pi}[\theta(\Lambda)-\theta(-\Lambda)], \tag{3}$$

$$E_{\rm mode}(\Lambda) = E_{\rm dens}(\Lambda) - B(\Lambda), \tag{4}$$

$$\Delta E_{\rm ren} = \lim_{\Lambda\to\infty}\left[E_{\rm mode}(\Lambda) + \frac{3k}{\pi}(\sqrt{1+\Lambda^2}-1)\right]. \tag{5}$$

The final line reproduces the existing closed energy formula; it does not change its finite part. The endpoint satisfies $B(\Lambda)/\Lambda \to 3k/(2\pi)$, so it cannot be omitted before applying the existing counterterm. The momentum endpoint $[q\theta(q)]_{-\Lambda}^{\Lambda}/(4\pi)$ tends to zero. Consequently the same conditional prescription gives

$$L_1(v) = v\Delta P_{\rm mode} - \Delta E_{\rm ren}. \tag{6}$$

This is **(c)** as a consequence of the stated spectral assumptions, rather than a completed physical determinant derivation. The original `charge_and_frame_audit.py:96--99` defines this combination; its quadrature at lines 220–245 checks that definition and the already subtracted energy integrand.

The equal-mode-count endpoint is also explicitly discussed by Faddeev and Korepin in Appendix 6, equations A.6.11–A.6.14. **(a)** Only the relevant full-text passages, rather than the whole paper, were read for this audit.

3 When differentiation gives the physical momentum

For an on-shell effective action with valid boundary variation, differentiation with respect to velocity computes the canonical charge. The first-order correction includes the classical charge evaluated on the corrected profile. It need not equal a fixed-background expectation alone. The finite-dimensional version makes this precise: let $L = L_0(v,y) + \hbar L_1(v,y)$, $L_{0,y}(v,y_0) = 0$, and $y = y_0 + \hbar y_1$. Assuming a nondegenerate Hessian after zero-mode projection, **(c)** (`audit_charges_action.py`)

$$y_0' = -\frac{L_{0,yv}}{L_{0,yy}}, \qquad y_1 = -\frac{L_{1,y}}{L_{0,yy}}, \qquad P_1 = L_{1,v} + L_{0,vy}y_1 = \frac{d}{dv}L_1(v,y_0(v)). \tag{7}$$

The field-theory application requires the corresponding functional inverse and boundary terms. In particular, changing the Boussinesq velocity also changes the asymptotic potential jump. The finite-dimensional identity does not check those boundary conditions. The physical identification of the published $\Delta P_{\rm can}$ is therefore conditional, although its derivative formula is correct. **(d)**

Taking (6) as input yields, independently at 65 digits, **(c)** (`audit_charges_action.py`)

$$\Delta P_{\rm can} = \frac{dL_1}{dv} = -\frac{\sqrt{3}\cos\alpha}{\pi}\log\left|\frac{\tan((\alpha+2\pi/3)/2)}{\tan((\alpha-2\pi/3)/2)}\right|, \quad (\epsilon=+1), \tag{8}$$

$$\Delta E_{\rm can} = v\Delta P_{\rm can} - L_1, \qquad \frac{d\Delta E_{\rm can}}{d\alpha} = v\frac{d\Delta P_{\rm can}}{d\alpha}. \tag{9}$$

These equations prove internal consistency, not a charge Ward identity for the renormalized field theory. Also, the printed intermediate equation for $-dL_1/d\alpha$ in the master note has the wrong overall sign: **(c)** (`audit_charges_action.py`)

$$\frac{dL_1}{d\alpha} = \frac{1}{2\pi}\sum_r c_r[\cos(2\phi_r) - \cos(\alpha+\phi_r)]\log|\tan(\phi_r/2)|. \tag{10}$$

The light-root cancellation survives this correction.

4 A static defect still has a localized momentum operator

Corrigan–Zambon give the conserved modified momentum, including localized field terms, in equations 3.11 and 3.49–3.51. It is not the ordinary translation Noether momentum of a translation-invariant system: a fixed defect breaks that symmetry. Their H contains a fifth-degree potential in the field jump (3.42–3.43), and their h is fourth-degree (3.46). **(a)** Relevant full-text sections 2–3 and 5.1 were read online, not the whole article. The dictionary is $U_{\mathrm{CZ}} = -3\phi_L$, $V_{\mathrm{CZ}} = -3\phi_R$, and $\mathcal{L}_{\mathrm{CZ}} = 9\mathcal{L}$; $c = \lambda = \kappa = 1$.

For direct independent verification, write $A = \eta/\sqrt{3}$, $B = 2A$, and denote a constant vacuum by σ. Expanding the boundary charges about that vacuum in project-normalized potential fluctuations a, b gives **(c)** (`audit_charges_action.py`, displayed for $\eta = +1$, with the same algebra for the other sign)

$$P_{\mathcal{D},2} = -\frac{A}{2}(a_x^2 + b_x^2) - Ba_xb_x - \frac{A}{4}(1-\sigma^2)(a-b)^2, \tag{11}$$

$$H_2 = \sigma\left[\frac{B}{2}(a_x^2 + b_x^2) + Aa_xb_x\right] + \frac{1-\sigma^2}{2}(a-b)(a_x+b_x) - \frac{B\sigma(1-\sigma^2)}{8}(a-b)^2, \qquad E_{\mathcal{D},2} = -H_2. \tag{12}$$

They are generally nonzero although the defect has no translational collective coordinate. A complete spectral trace of the conserved charge might already include these terms through its mode normalization. This audit does not assert it cannot. What is absent from the published check is a derivation equating that trace to $\int q\theta_D'(q)\,dq/(4\pi)$. Staticity alone cannot supply that derivation. **(d)**

5 The defect subtraction requires a boundary derivation

For the existing two-root kernel, put $\cos\gamma = \sigma/2$, $\psi_\pm = \pm 2\pi/3 - \gamma$ and weights $(-\eta, +\eta)$. Then **(c)** (`audit_charges_action.py`)

$$S_D := \sum c_D \sin\psi = -\eta\sqrt{3}\cos\gamma = -\frac{\eta\sqrt{3}\sigma}{2}, \qquad -\frac{\omega'(q)[\theta_D(q) - \theta_D(-q)]}{4\pi} \longrightarrow \frac{S_D}{\pi} \quad (q \to +\infty). \tag{13}$$

For example $\eta = +1$, $\sigma = -1.5$ gives slope $0.413496671566\ldots$. This is a nonzero continuum divergence even on a linearly stable vacuum.

The existing bulk normal-ordering counterterm is proportional to the sum of $\int \phi_x\,dx$ over the two open half-lines. Each integrand vanishes on a constant defect vacuum. Hence that bulk term cannot by itself cancel the displayed divergence. Interpreting the jump as a distribution at the defect would introduce a boundary prescription which must be derived and specified. The statement in `oneloop_defect_phase.tex:611--613` that the potential is linear is contradicted by its fifth-degree term. Thus the finite defect energy formula is an algebraically coherent subtraction, but its physical renormalization argument remains unestablished. **(d)**No replacement prescription is adopted here.

6 A conditional absorption-locus shift

Suppose both that the canonical soliton charges are physical and that the quoted defect vacuum shifts are physical, while retaining their algebraic mode-charge jump identity. Conservation would then require more than the published assertion that static absorption uses a different bulk momentum. There is a simple kinematic possibility: the absorption soliton could have $\alpha_{\mathrm{abs}} = \alpha_0 + \hbar a_1 + O(\hbar^2)$. Because **(c)** (`audit_charges_action.py`)

$$\Delta E_{\mathrm{can}} - \Delta E_{\mathrm{ren}} = v(\Delta P_{\mathrm{can}} - \Delta P_{\mathrm{mode}}), \qquad E_{\mathrm{cl}}' = vP_{\mathrm{cl}}', \tag{14}$$

the choice

$$a_1 = \frac{\Delta P_{\text{mode}} - \Delta P_{\text{can}}}{P'_{\text{cl}}} \tag{15}$$

makes both first-order canonical soliton charges equal the mode-charge increments, whenever $P'_{\text{cl}} \neq 0$. This is a conditional algebraic observation **(c)**, not a derived quantum absorption condition **(d)**. At $\alpha_0 = 0.15, 0.5, 0.85$ the coefficients are respectively 0.753417205691, 0.920879681850, and 2.33030395807. Quantum shifts of defect parameters or additional localized charge terms are also not excluded. Selecting among these requires a renormalized quantum sewing or charge calculation, not a choice made in this audit.

7 Checks, limitations, and failures recorded

Claim checked	Verification
Co-moving quadratic action	Exact SymPy Taylor coefficient
Existing bulk energy with cutoff endpoint	65-digit closed integration; cutoff 10^{20}, residual below 10^{-19}
Legendre formulas and absorption-shift identities	Derivatives checked to 10^{-59} or better at three regular light solitons
Stationary profile correction	Exact symbolic chain-rule identity
Localized quadratic defect charges	Independent symbolic expansion
Defect UV slope	Exact trigonometry and 65-digit large-q check

Failures recorded, including statements checked and found false:

1. The master note's profile at line 250 is $-k^2 \operatorname{sech}^2(k\xi/2)$, whereas the fixed convention gives $+k^2 \operatorname{sech}^2(k\xi/2)/2$. The displayed profile fails the travelling-wave equation already at its centre (checked symbolically).

2. The weighted cosine sum stated to vanish at line 257 is actually $\sum c_r \cos\phi_r = -3\cos\alpha$. The final charge derivative can still be correct because cancellations occur within the energy derivative.

3. The sign of the printed $-dL_1/d\alpha$ weight formula fails; the same weight equals $+dL_1/d\alpha$. The closed canonical momentum passes.

4. The defect potential is not linear. Its fifth-degree term and its localized quadratic momentum contradict the reasons given for discarding boundary effects.

5. No implementation assertion failed during development of this audit script; no tolerance was weakened. A direct DOI webpage lookup for Faddeev–Korepin was inaccessible; the DOI was subsequently verified against the publisher's indexed entry.

The full determinant, projected measure, quantum boundary counterterms, and quantum charge Ward identities were not computed. The successful script therefore ends with `ALL CHECKS PASSED` only for its stated algebraic and conditional checks, not for the unresolved physical identifications.

References

[1] E. Corrigan and C. Zambon, *Adding integrable defects to the Boussinesq equation*, arXiv:2306.03772. Relevant full-text sections 2–3 and 5.1 read online for this audit.

[2] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Physics Reports **42** (1978), 1–87, doi:10.1016/0370-1573(78)90058-3. DOI verified against the publisher's indexed entry; reading scope stated above.